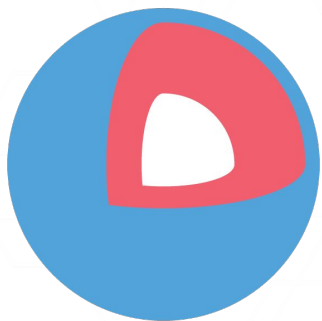




End-to-end Monitoring with the Prometheus Operator

By @mxinden

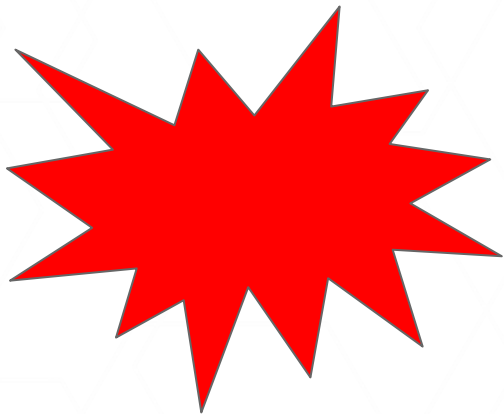
Secure, simplify and automate container infrastructure



Core OS

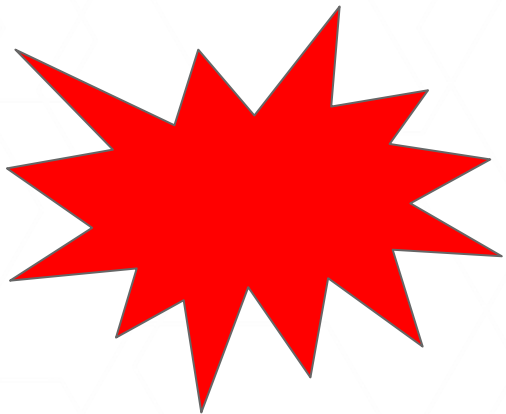
Why Monitoring?

Why Monitoring?

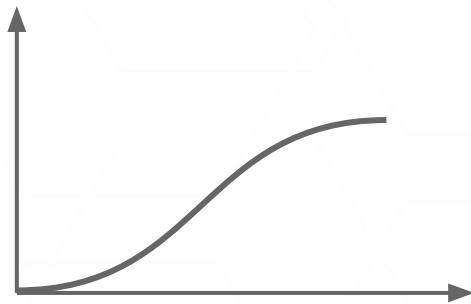


Alerting

Why Monitoring?



Alerting



Long-term trends

What is Prometheus?



- Open Source Monitoring
- Built by Soundcloud
- Inspired by borgmon

What is Prometheus?



- Pull-based

What is Prometheus?



- Pull-based
- Multi-Dimensional

What is Prometheus?



- Pull-based
- Multi-Dimensional
- ***Metrics***, **not** logging, **not** tracing

What is Prometheus?

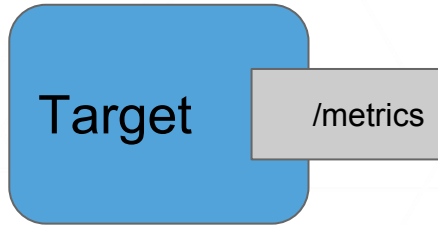
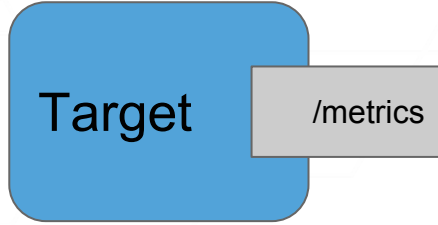
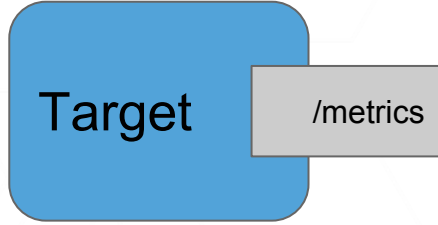


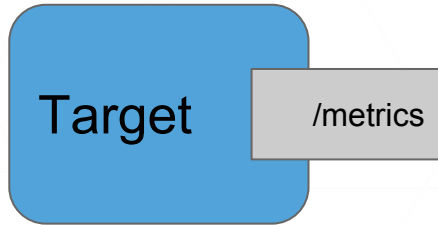
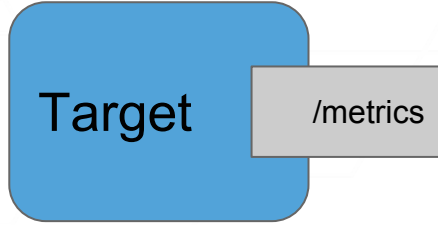
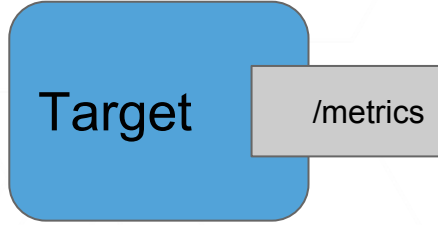
- Pull-based
- Multi-Dimensional
- ***Metrics***, **not** logging, **not** tracing
- No magic!

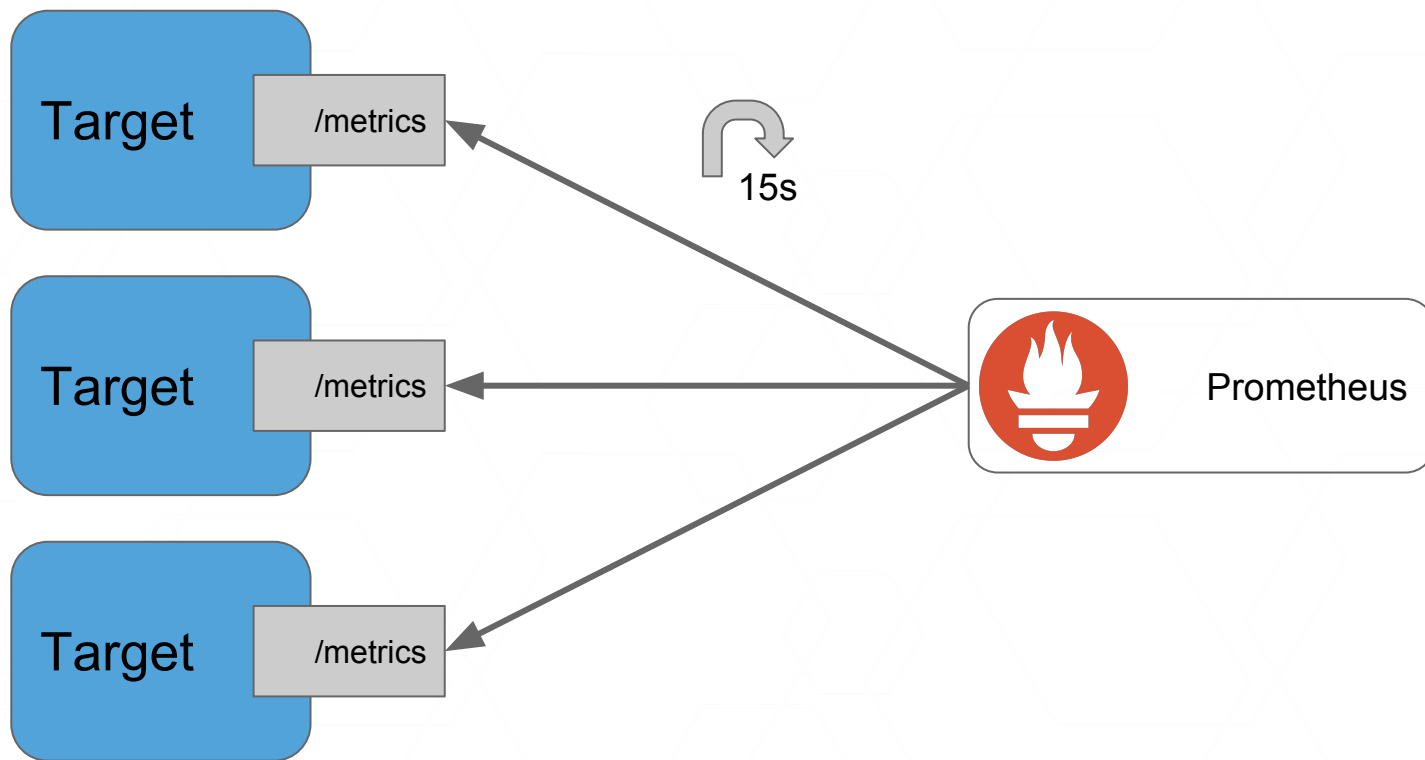
Target

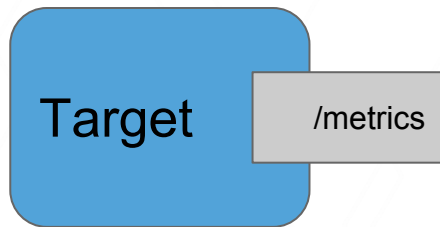
Target

Target

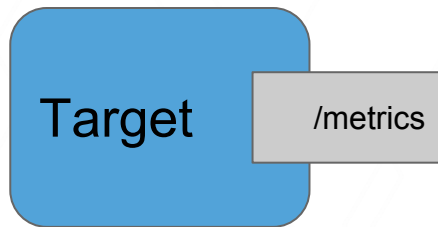








```
# HELP http_requests_total Total number of HTTP requests made.  
# TYPE http_requests_total counter  
http_requests_total{code="200",path="/status"} 8
```

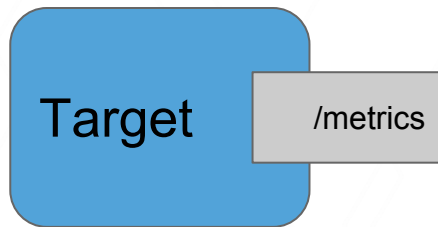


```
# HELP http_requests_total Total number of HTTP requests made.
```

```
# TYPE http_requests_total counter
```

```
http_requests_total{code="200",path="/status"} 8
```

Metric name

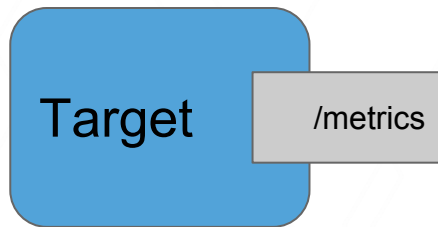


```
# HELP http_requests_total Total number of HTTP requests made.
```

```
# TYPE http_requests_total counter
```

```
http_requests_total{code="200",path="/status"} 8
```

Label

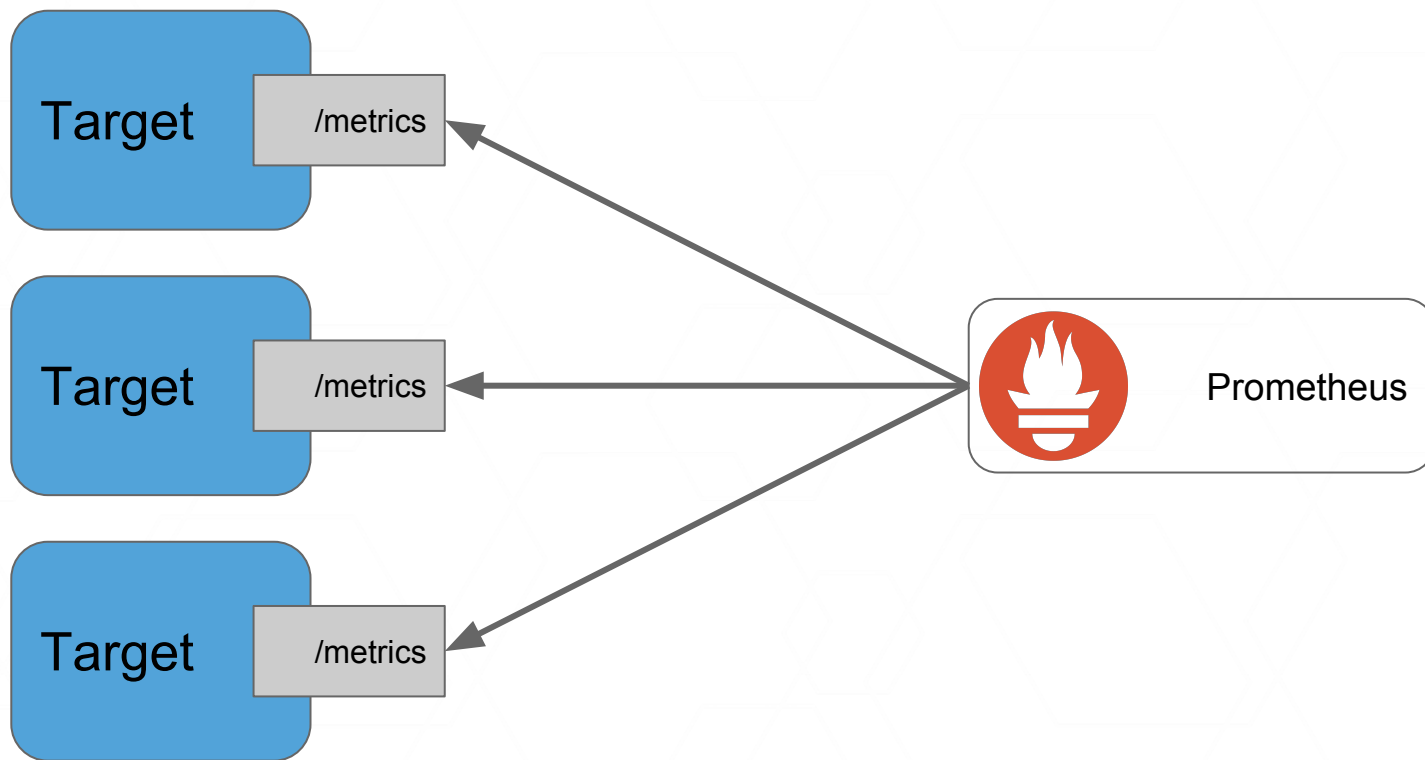


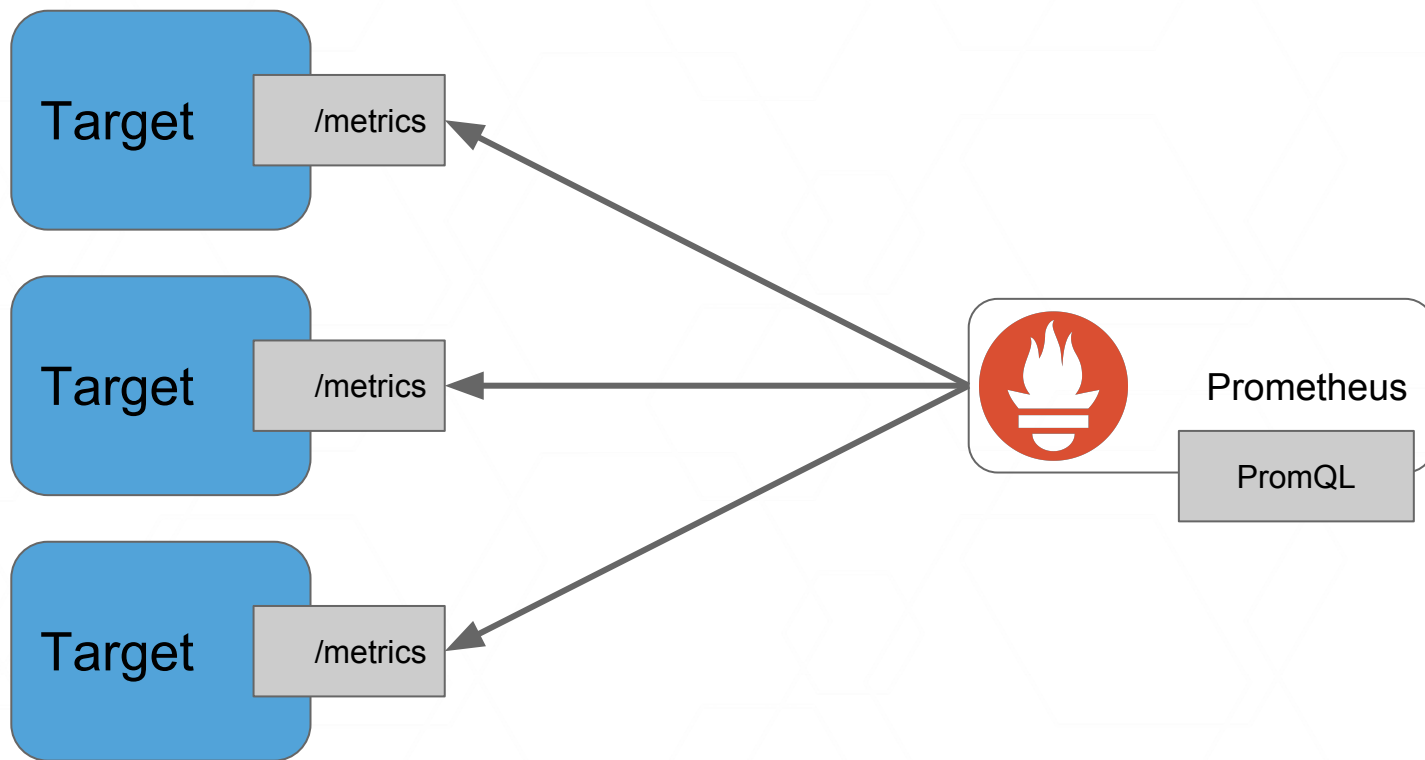
```
# HELP http_requests_total Total number of HTTP requests made.
```

```
# TYPE http_requests_total counter
```

```
http_requests_total{code="200",path="/status"} 8
```

Value





Current percentage of HTTP errors across all service instances?

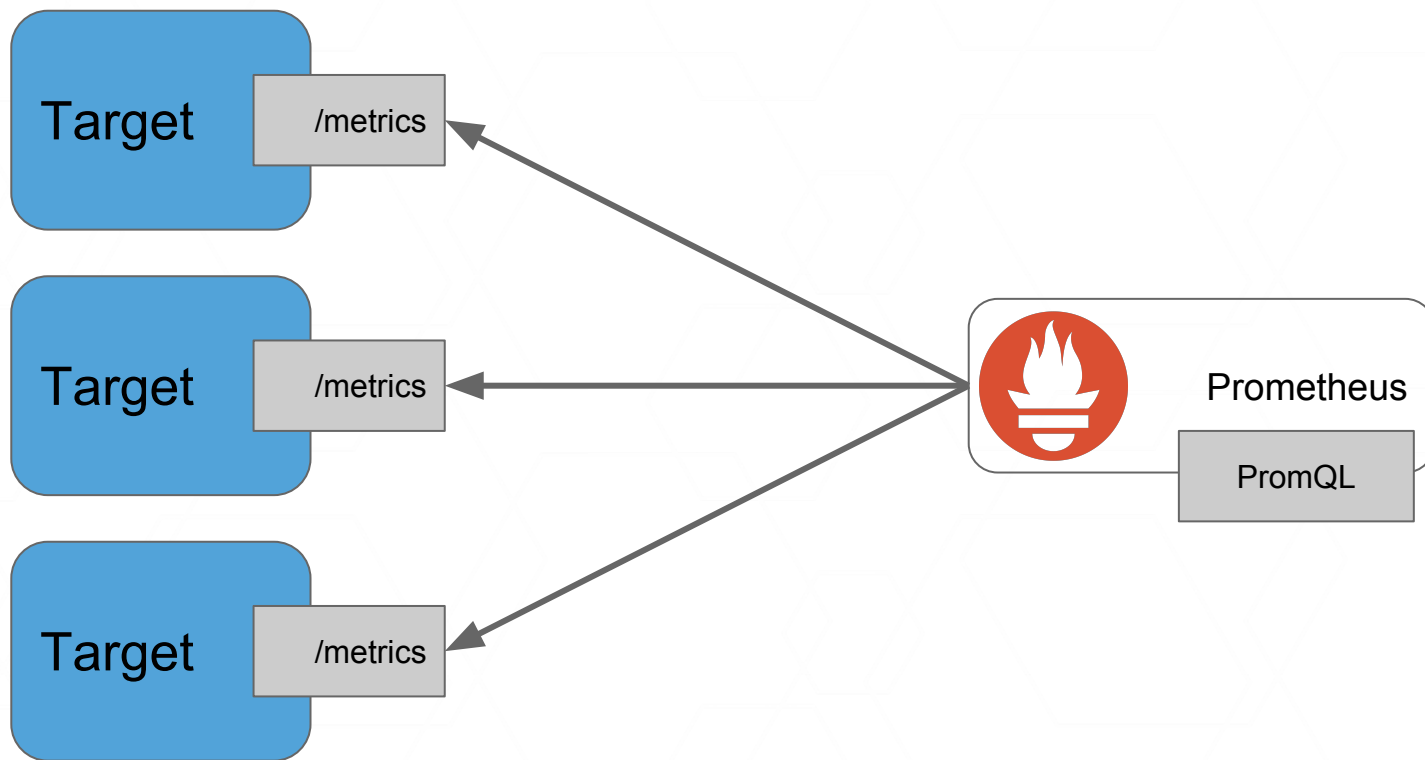
Current percentage of HTTP errors across all service instances?

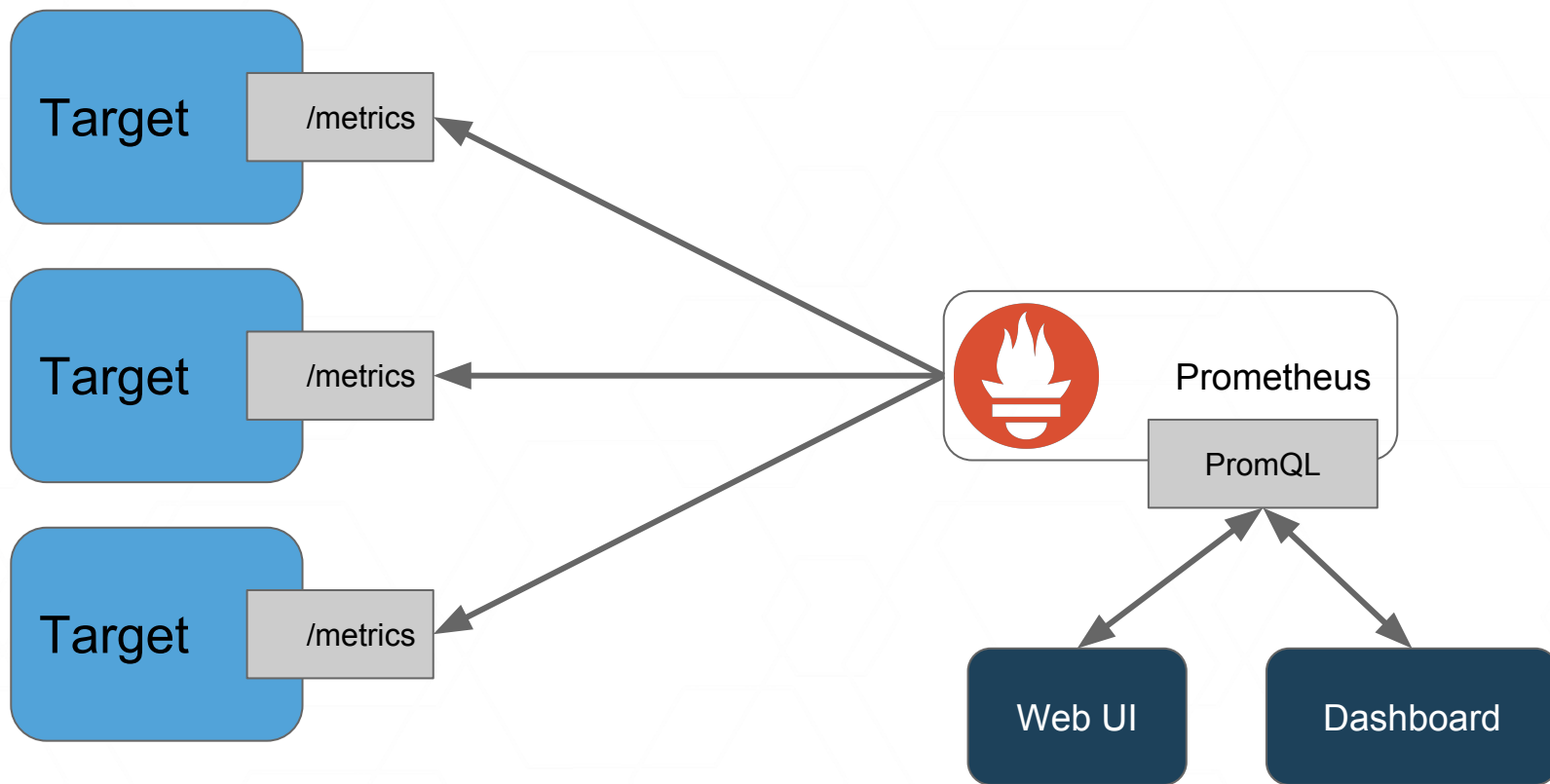
```
sum by(path) rate(http_requests_total{status="500"}[5m]))  
/ sum by(path) rate(http_requests_total[5m]))
```

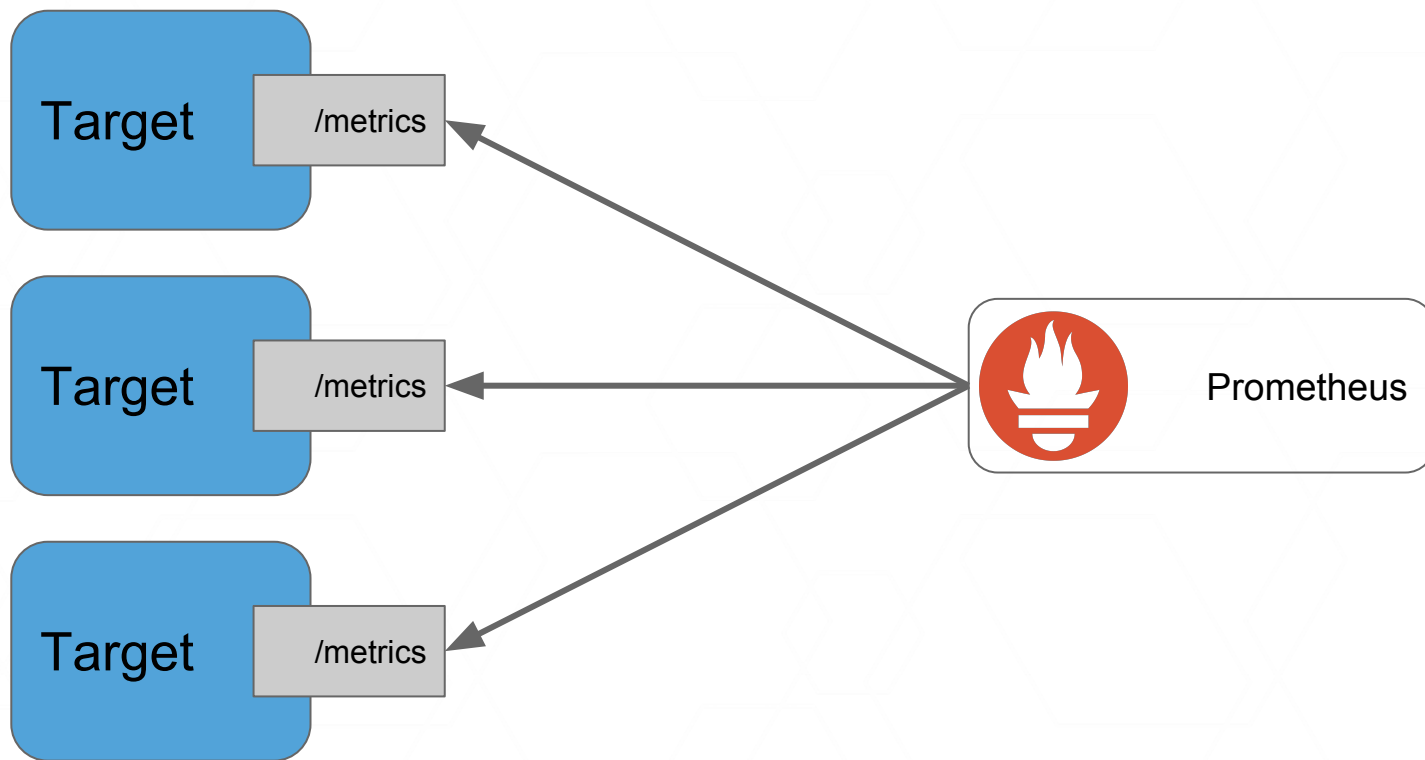
Current percentage of HTTP errors across all service instances?

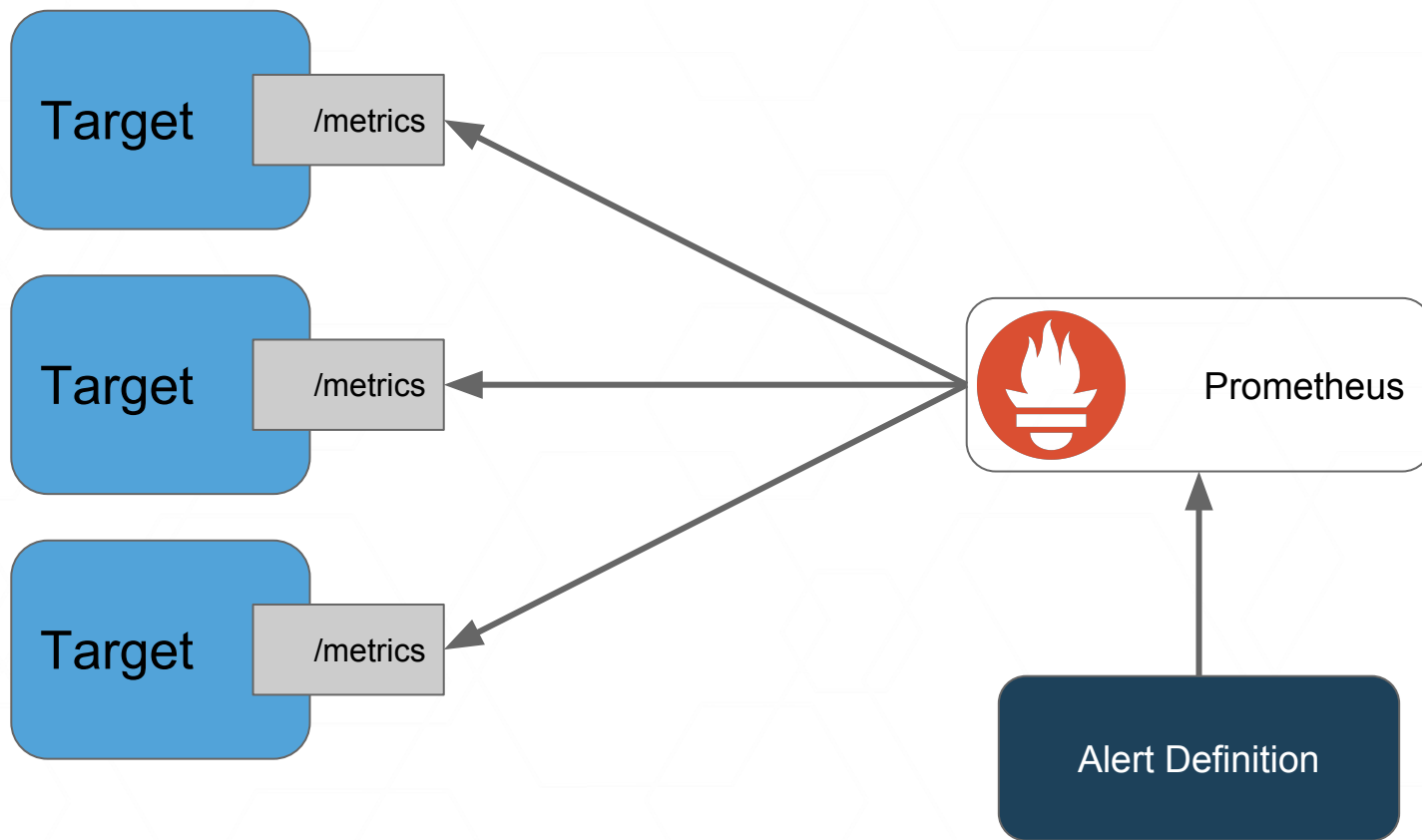
```
sum by(path) rate(http_requests_total{status="500"}[5m]))  
/ sum by(path) rate(http_requests_total[5m]))
```

{path="/status"}	0.0039
{path="/"}	0.0011
{path="/api/v1/topics/:topic"}	0.087
{path="/api/v1/topics"}	0.0342





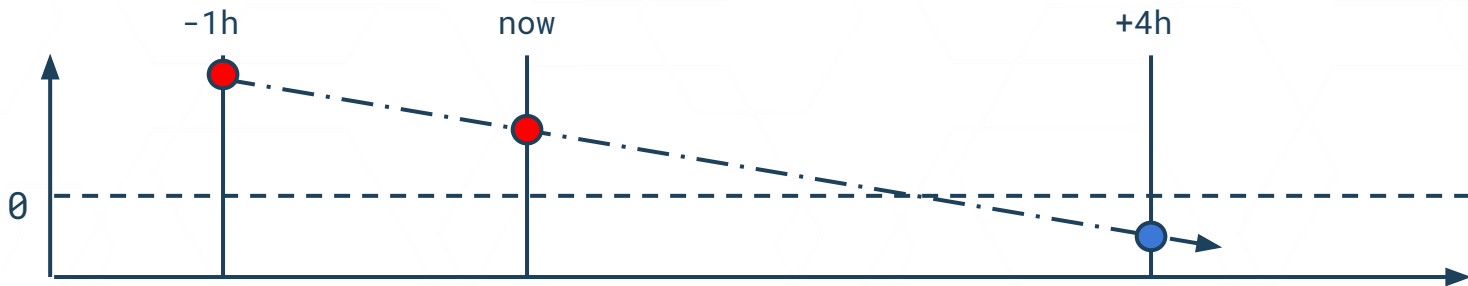


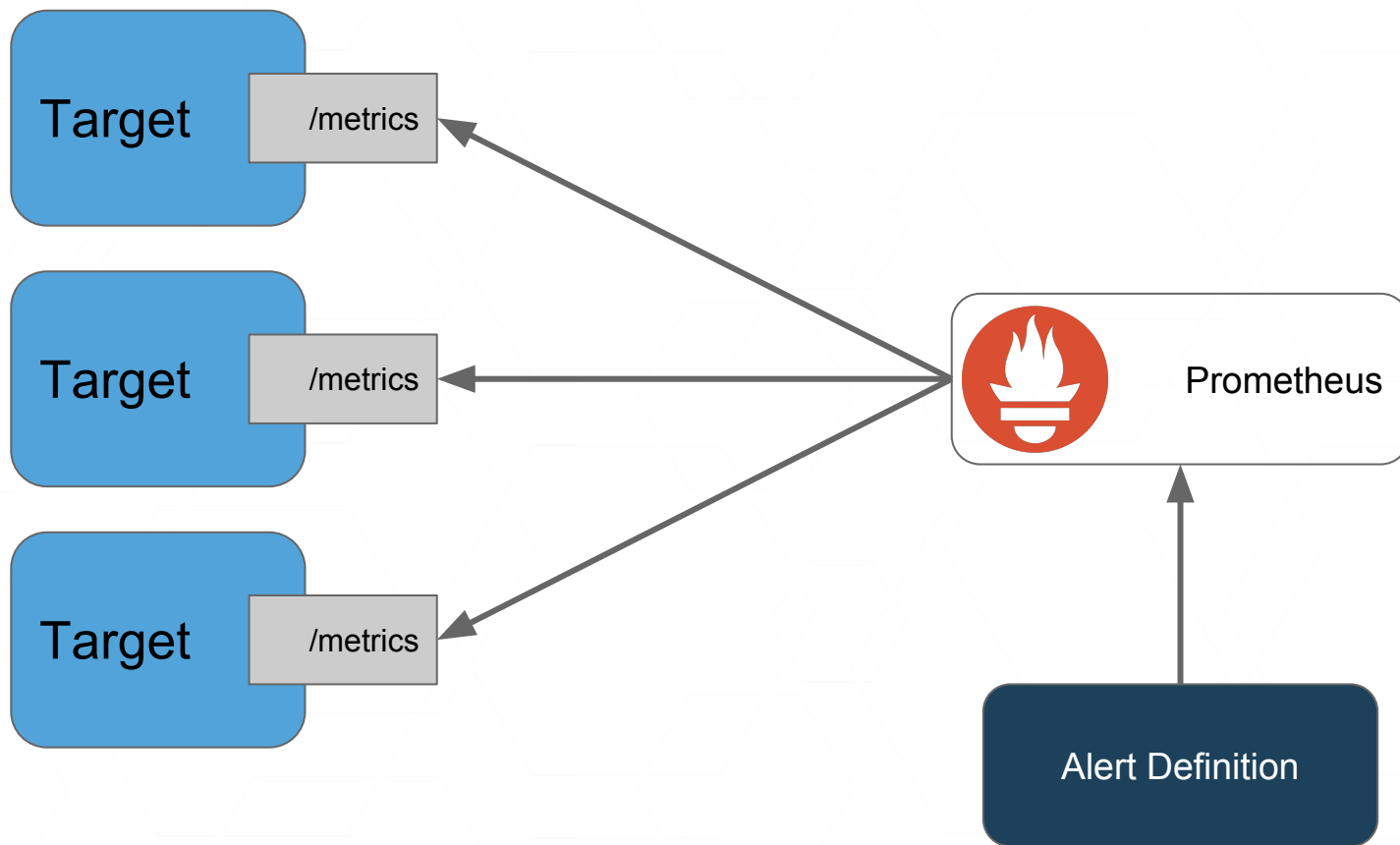


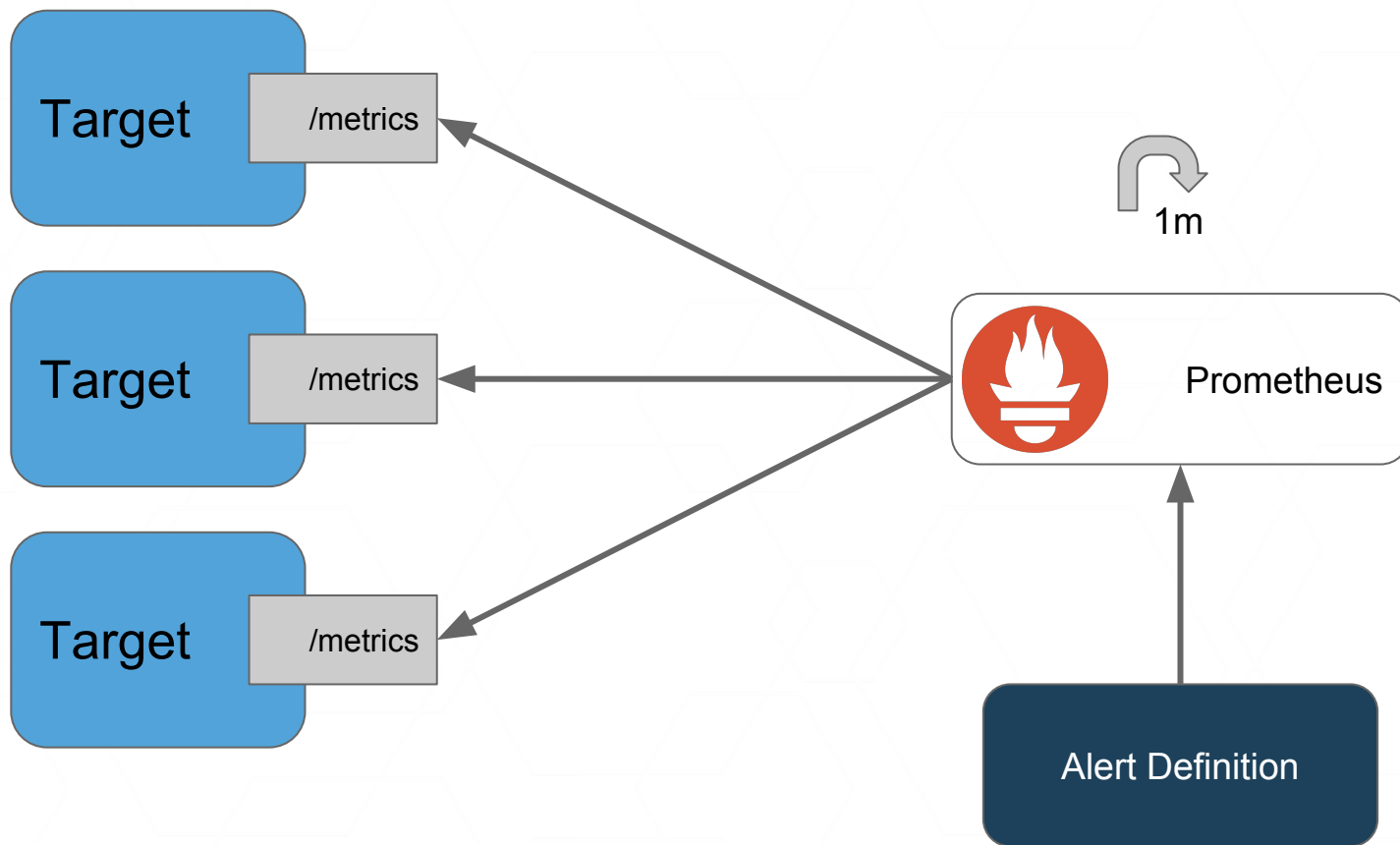
Is any disk about to run full within 4 hours?

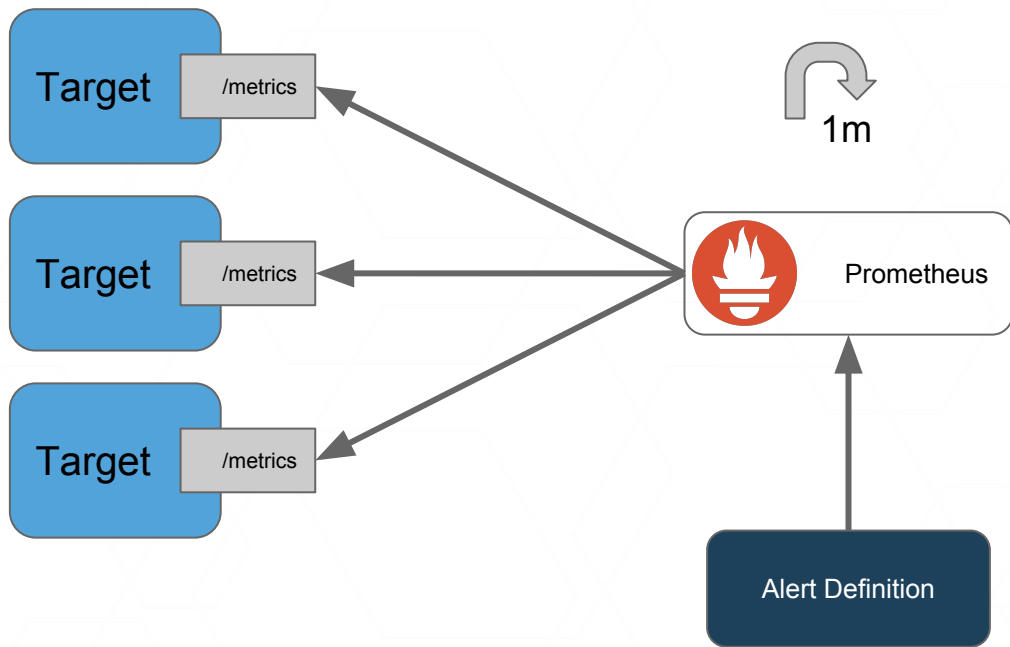
ALERT DiskWillFillIn4Hours

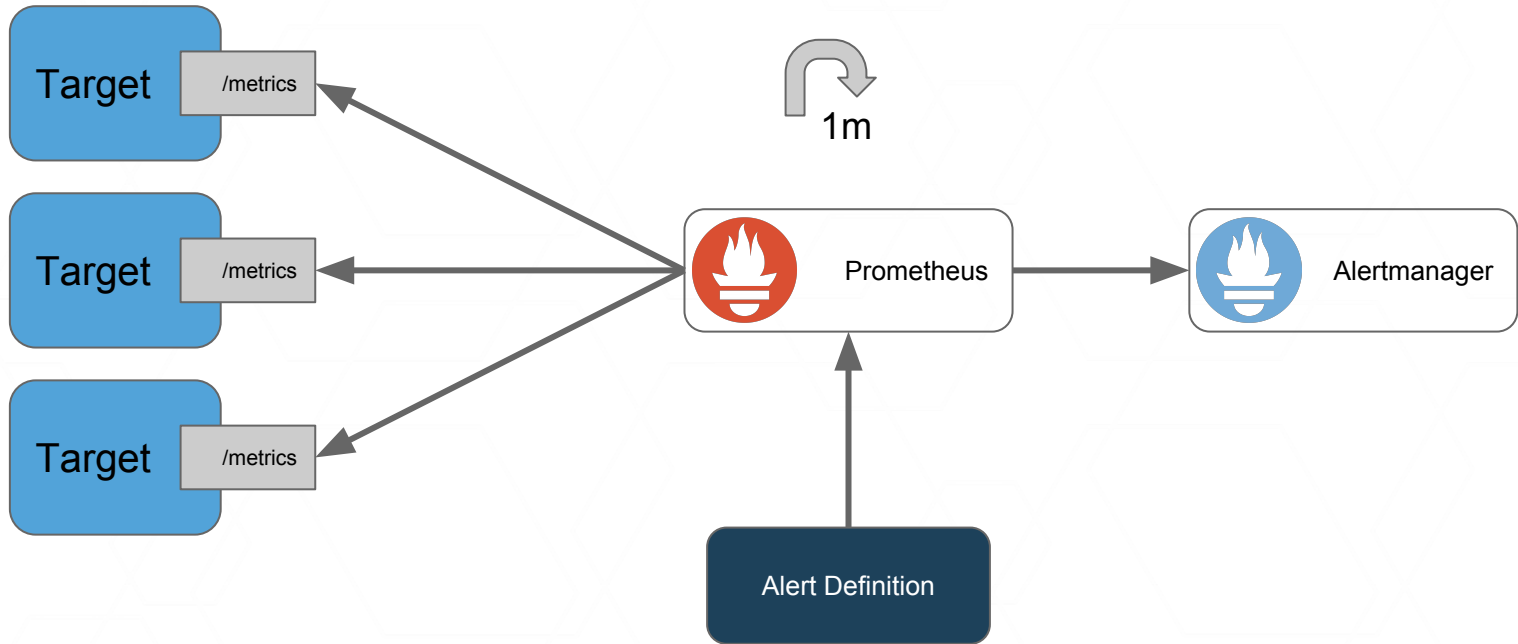
IF `predict_linear(node_filesystem_free[1h], 4*3600) < 0`





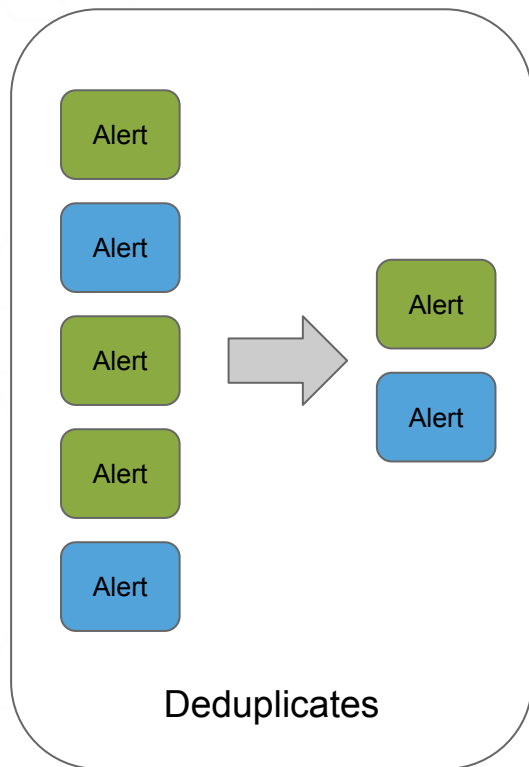






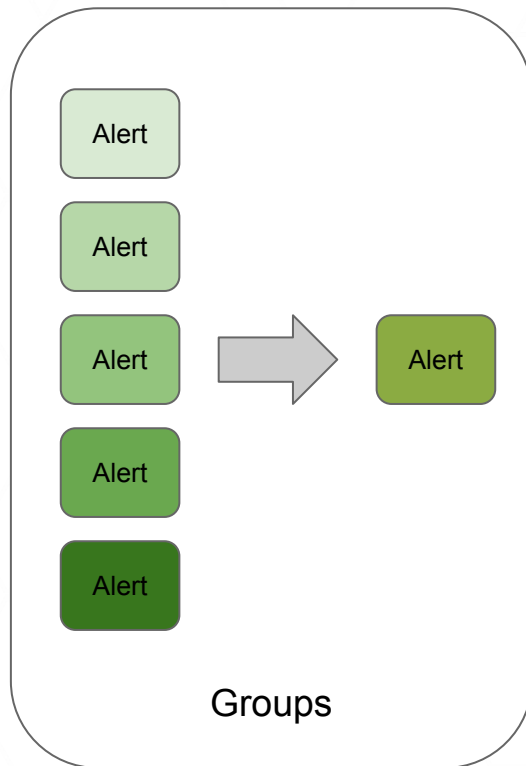
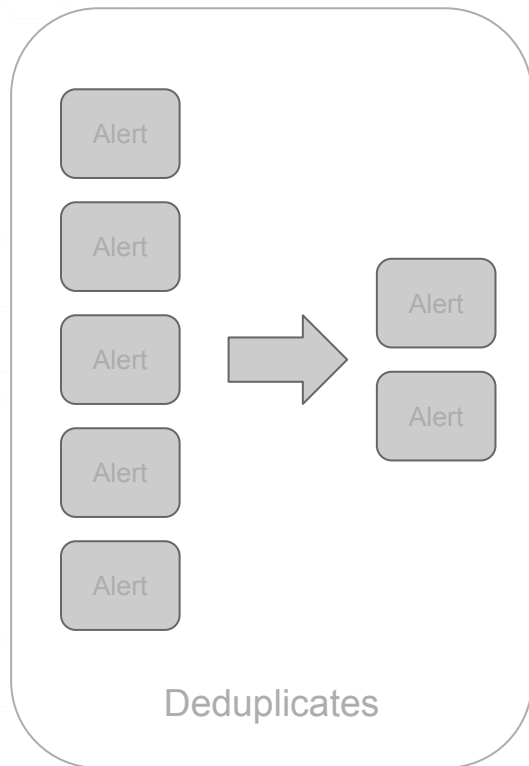


Alertmanager



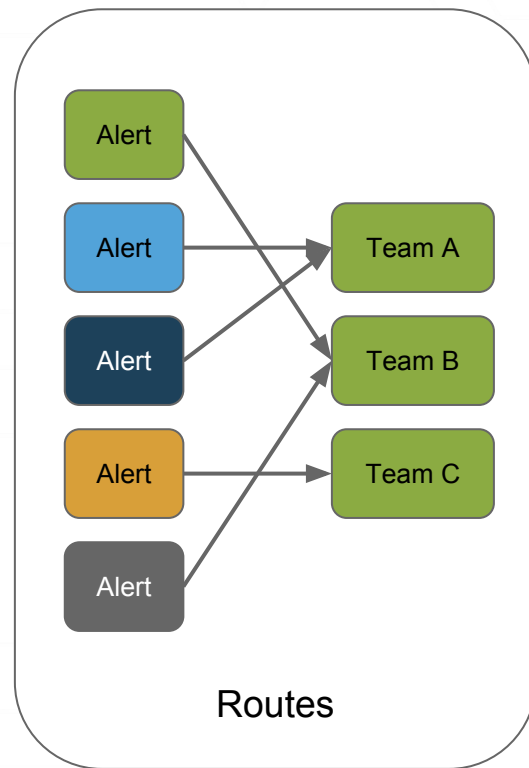
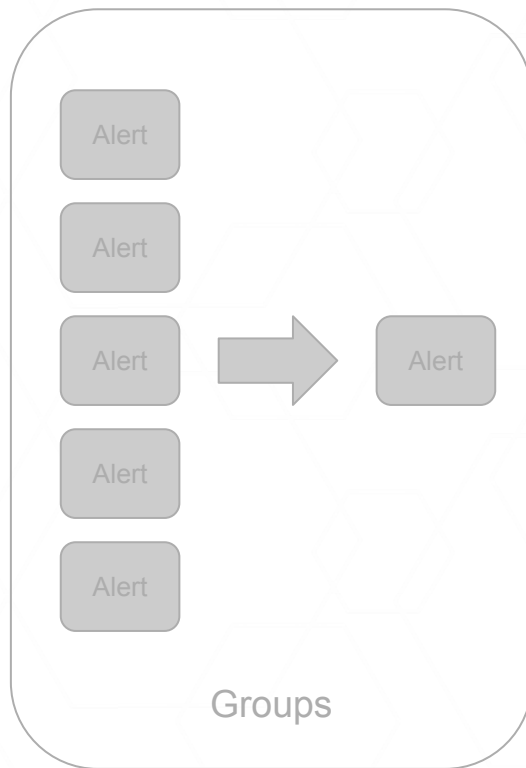
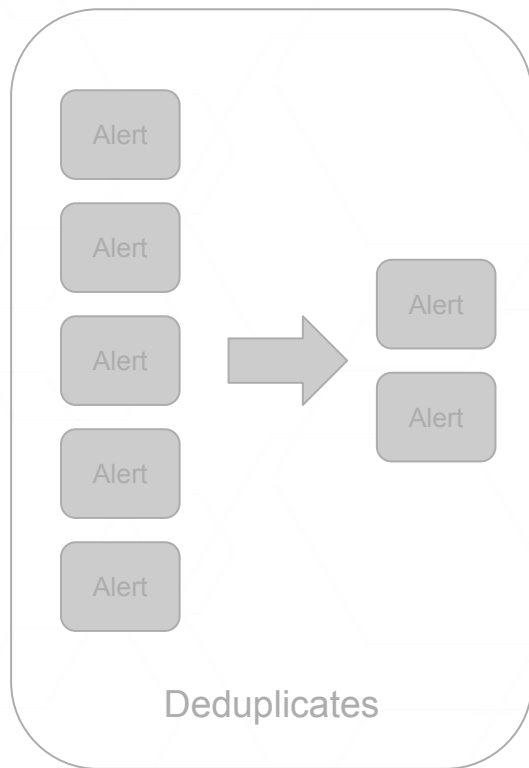


Alertmanager



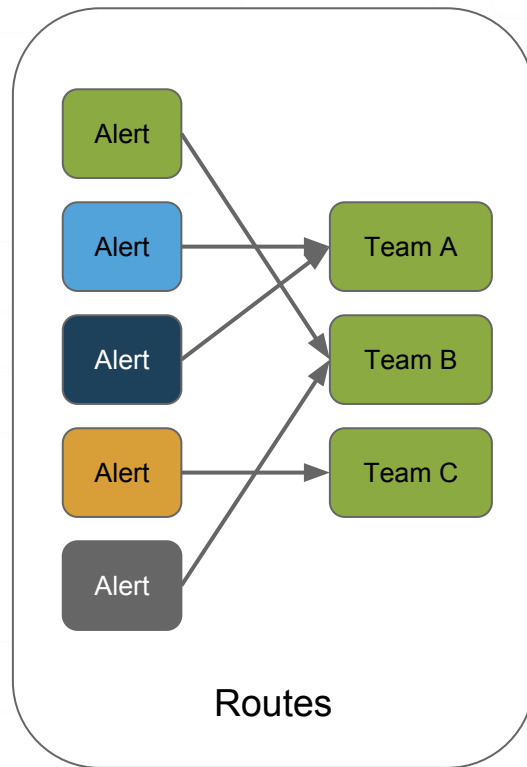
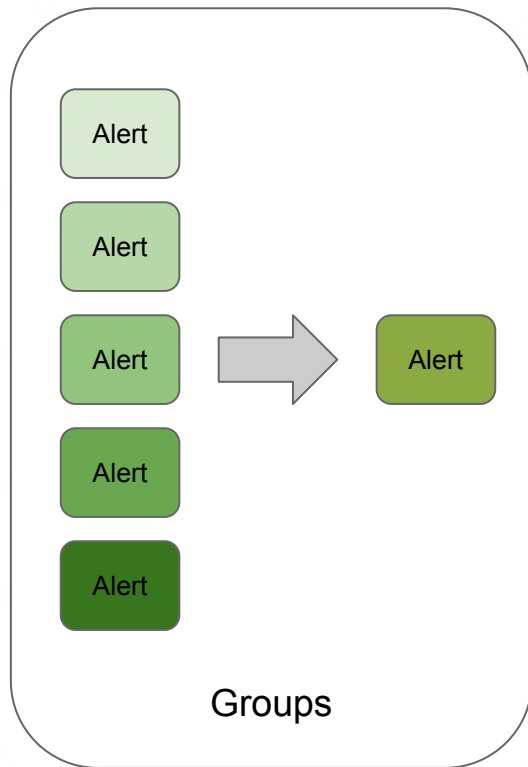
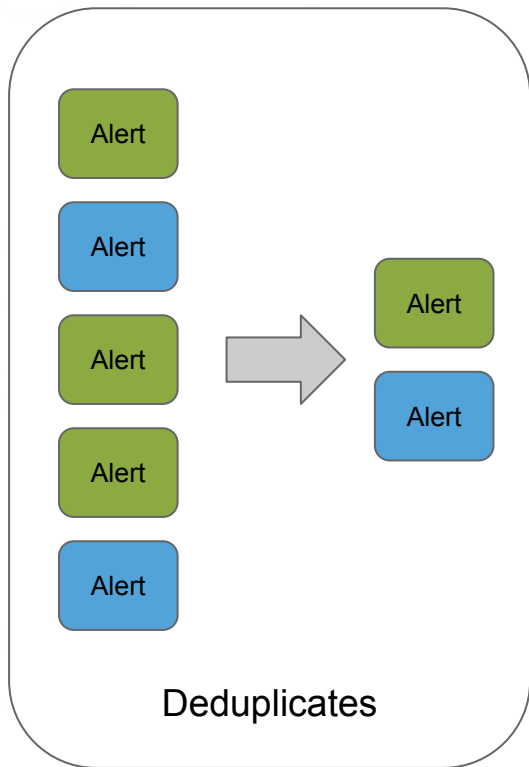


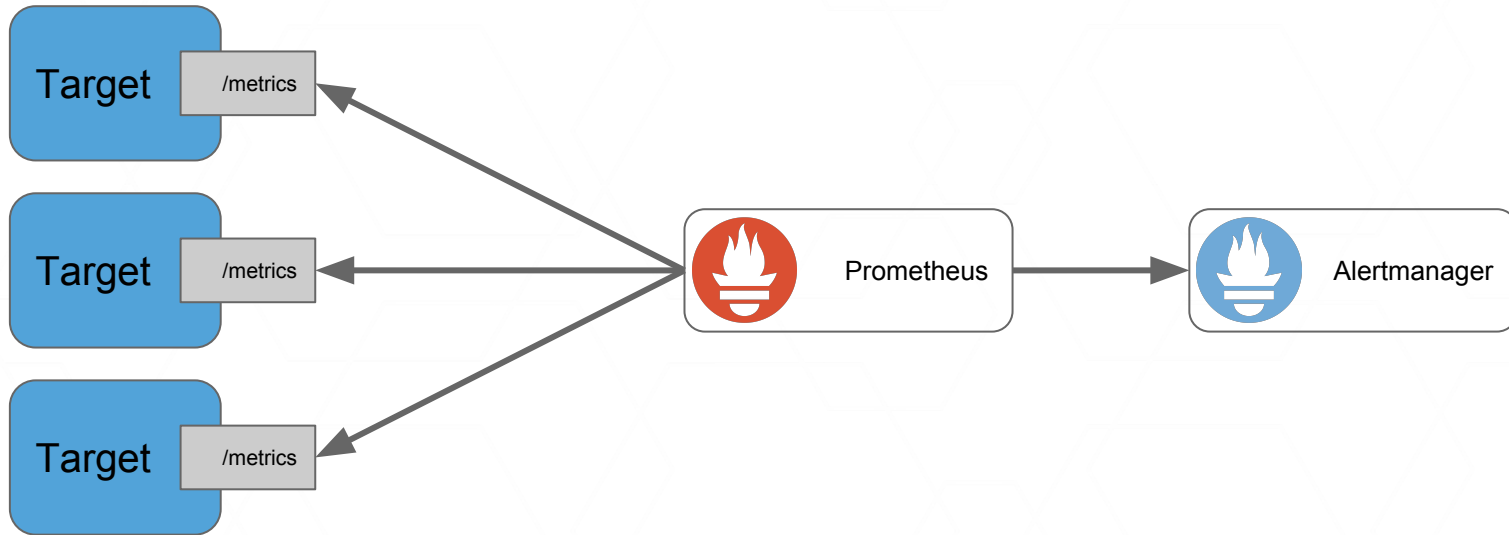
Alertmanager

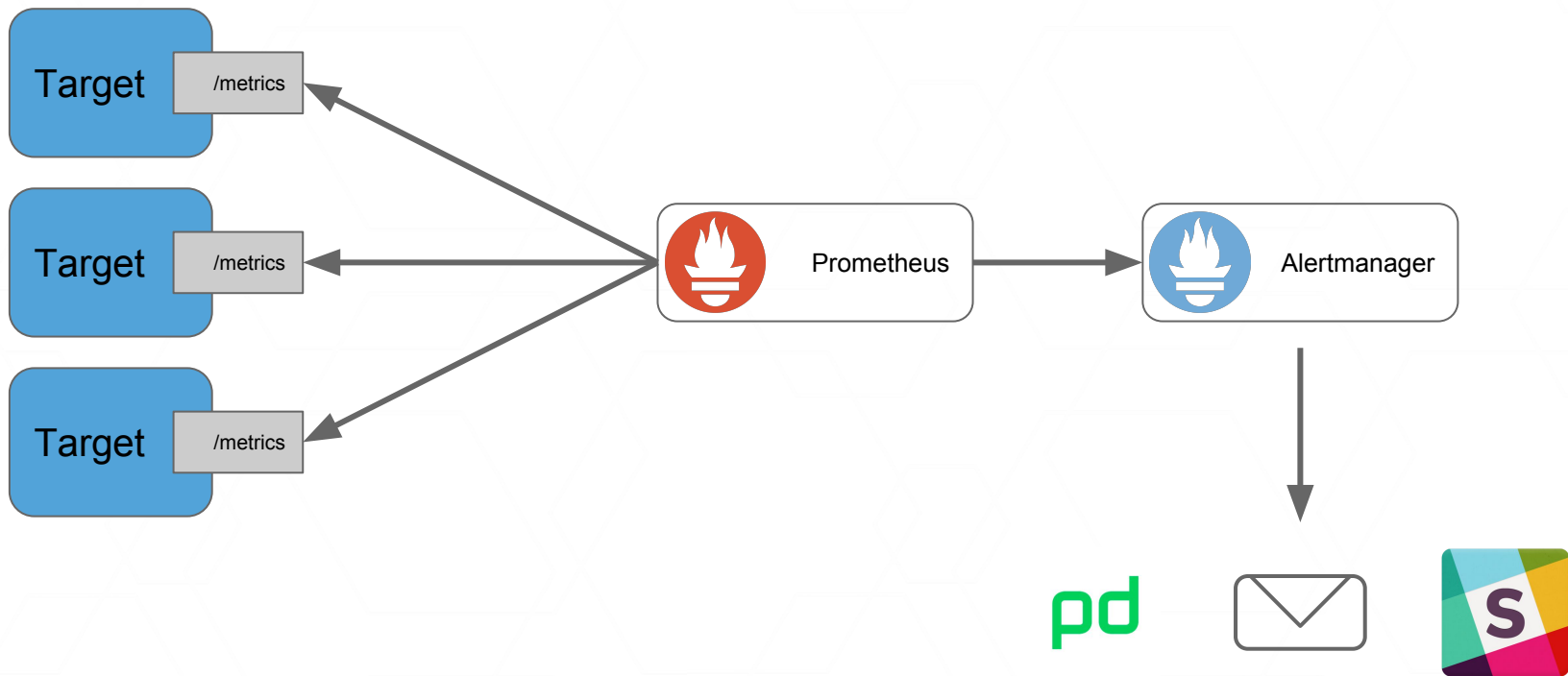




Alertmanager

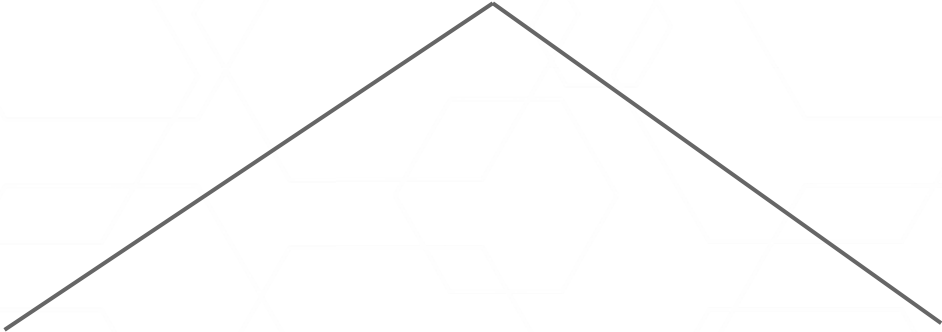






Monitoring

Monitoring



```
graph TD; Monitoring --> Application; Monitoring --> Cluster;
```

Application

Cluster

Cluster Monitoring

What is Kubernetes?



Platform for running
containerized applications

What is Kubernetes?



Announced 2014 by *Google*
Influenced by *Borg & Omega*

v1.01 in July 2015
Kubernetes joins the CNCF

Master

Master

API-Server

etcd

Controller-Manager

Scheduler

Kube-DNS

...

Master

API-Server

etcd

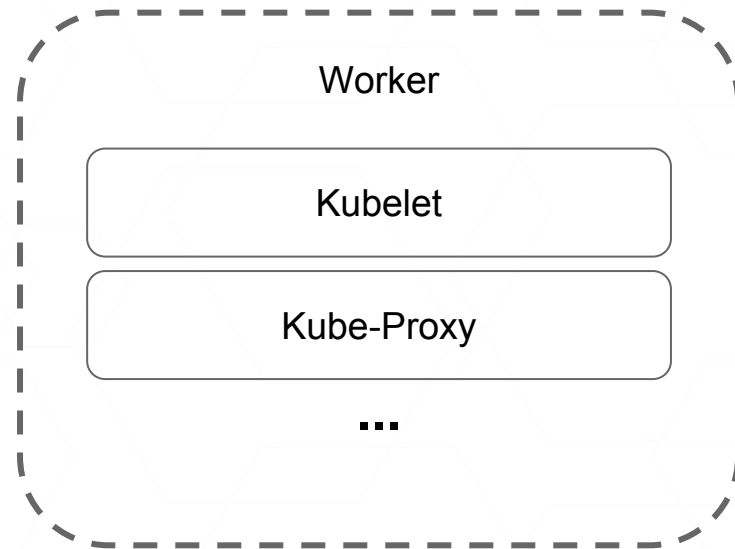
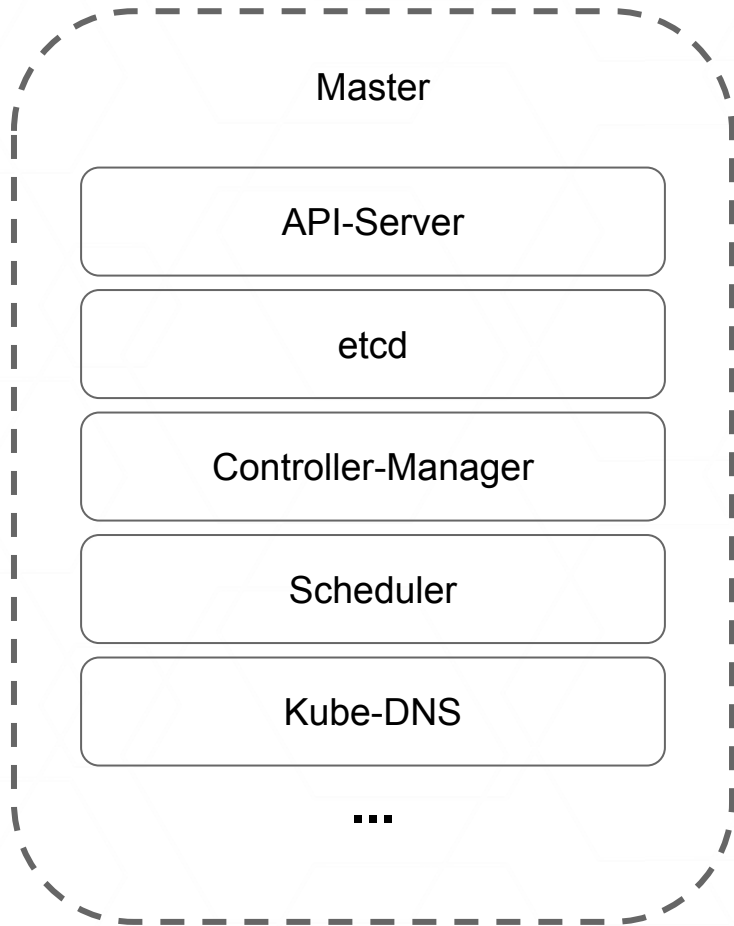
Controller-Manager

Scheduler

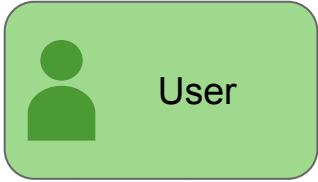
Kube-DNS

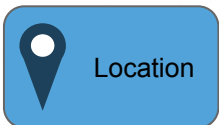
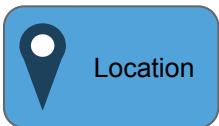
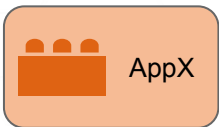
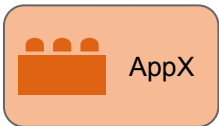
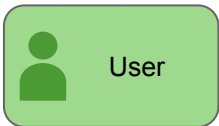
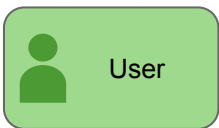
...

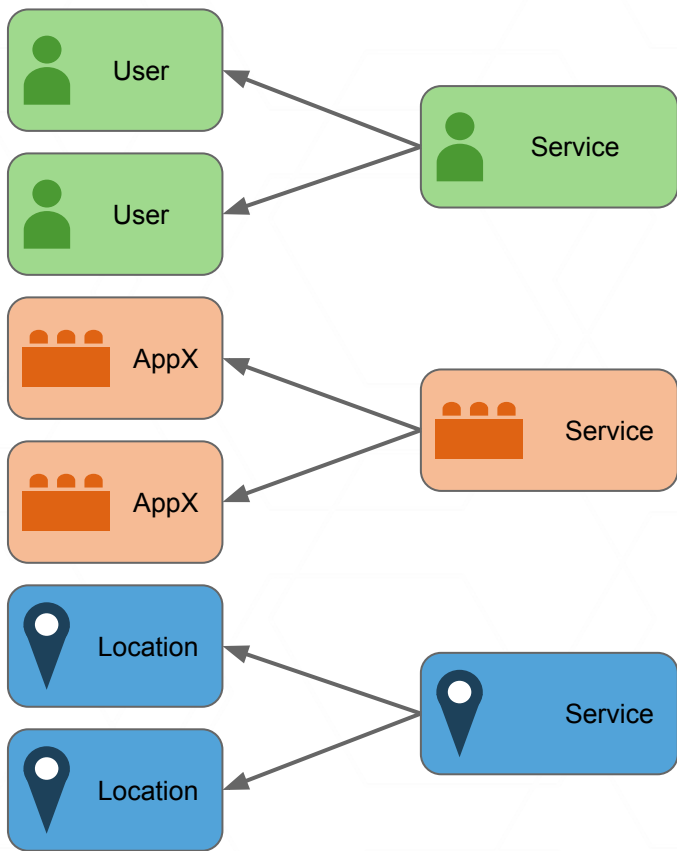
Worker

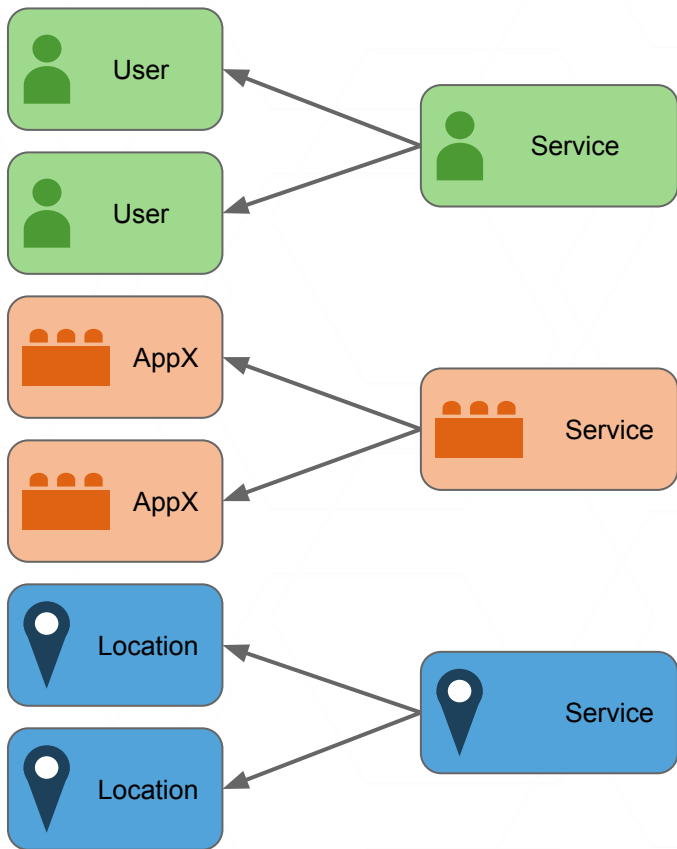


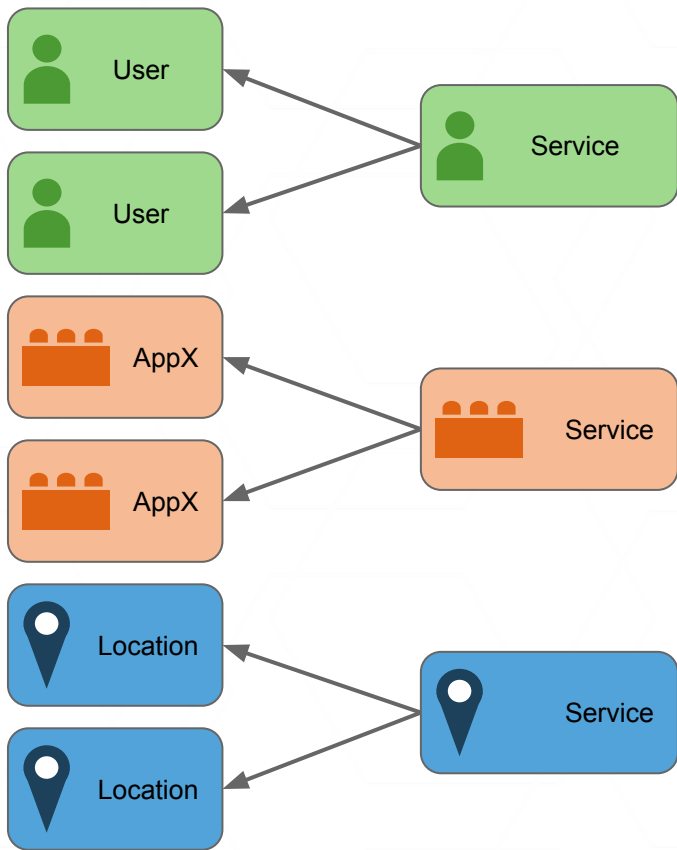
Application Monitoring





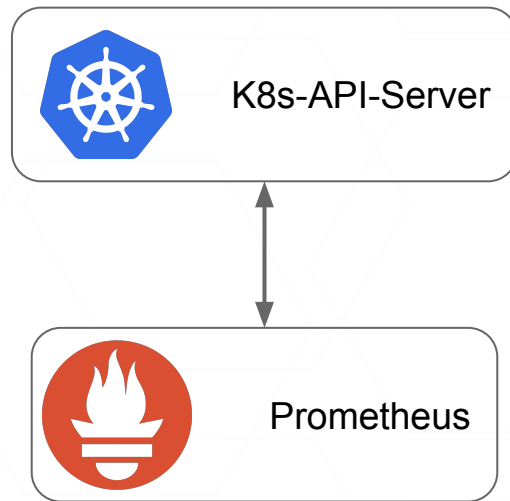
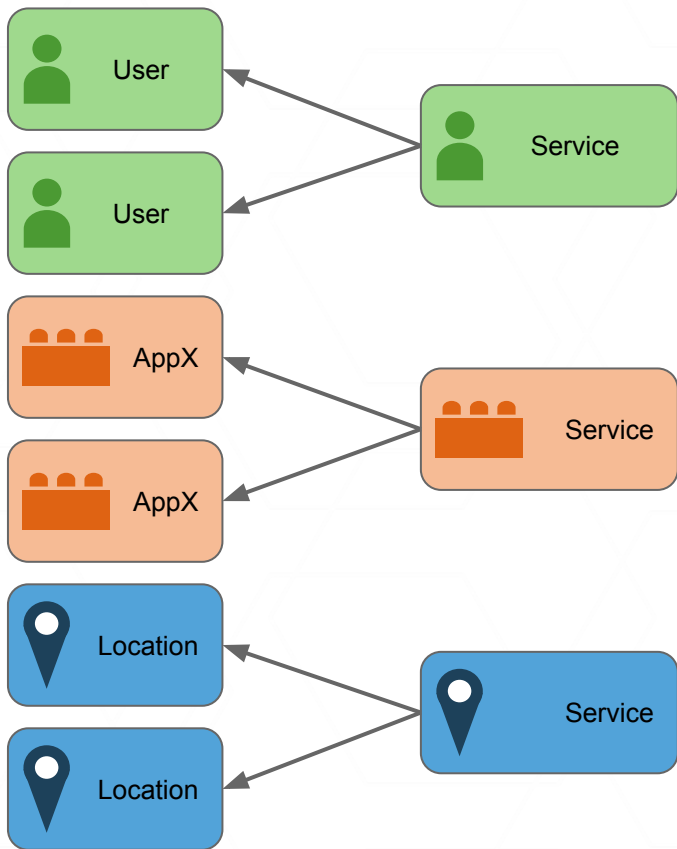


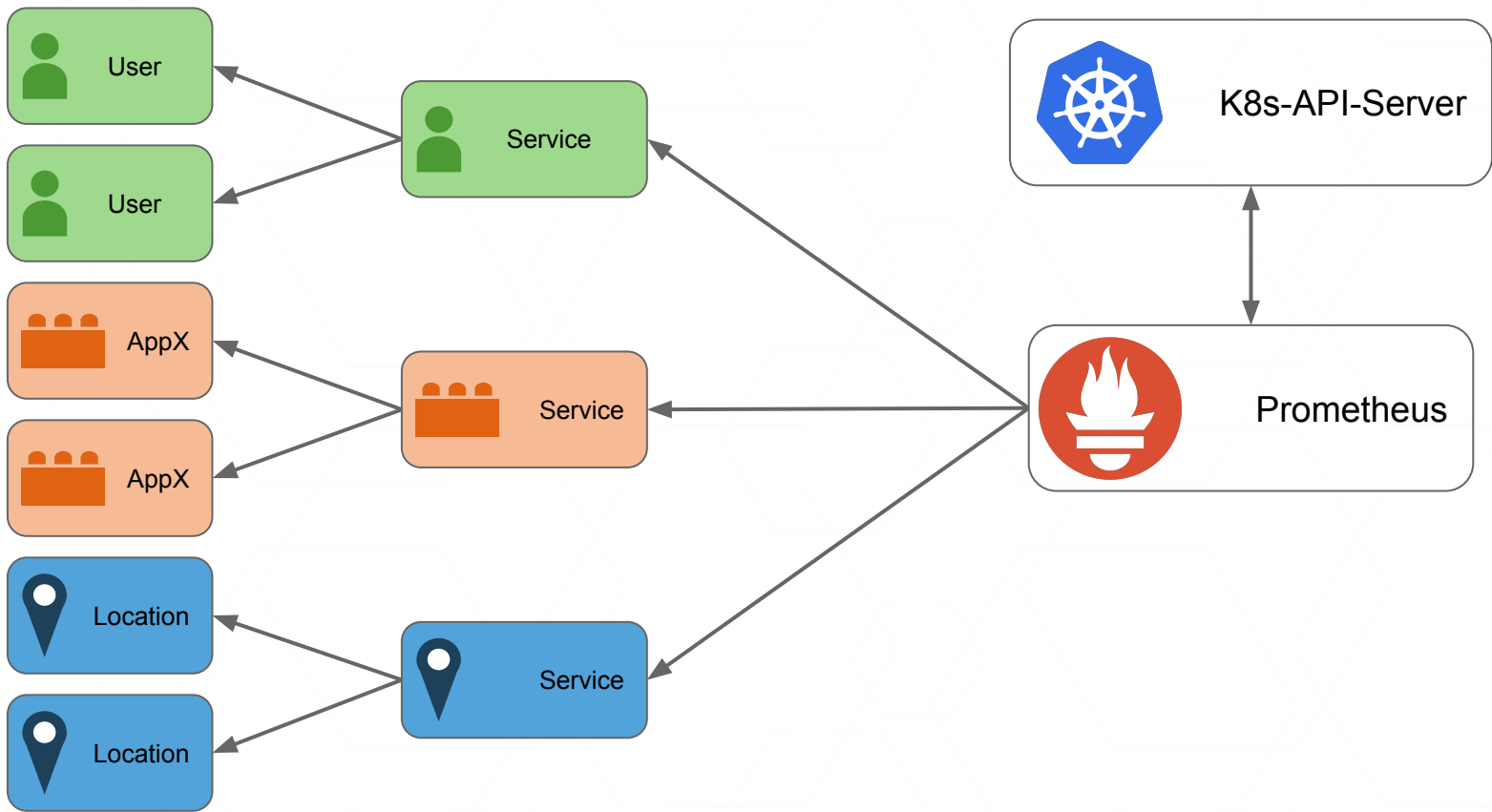




?

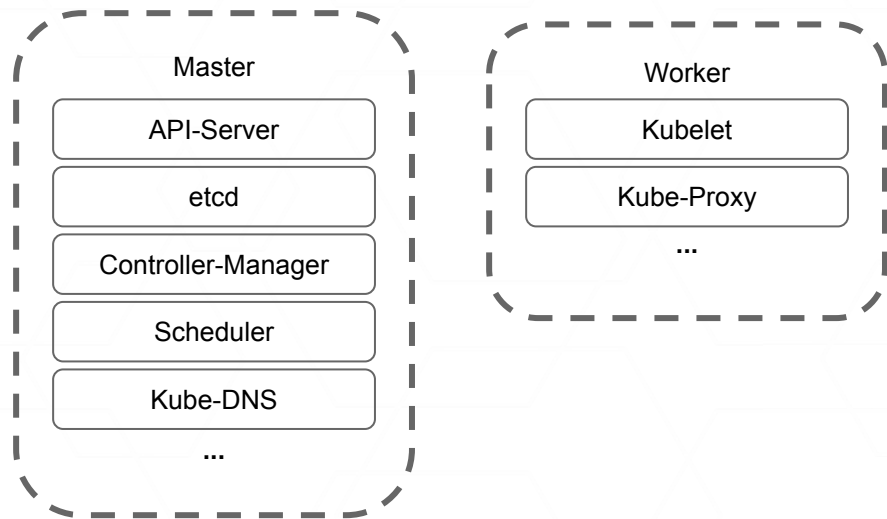




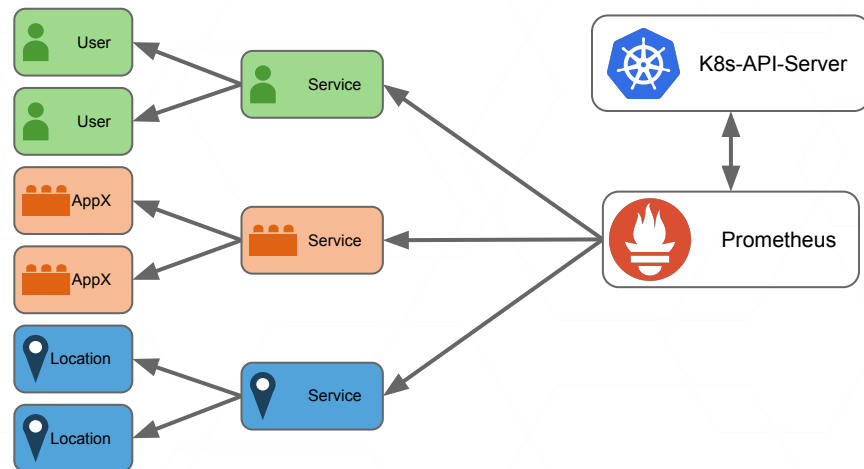


Service Discovery

- Static target list
- DNS discovery
- Kubernetes discovery
- ...



Cluster-Monitoring



Application-Monitoring

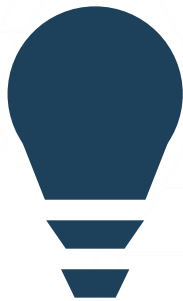
Problem

Prometheus is **stateful** and **difficult to configure!**

Introducing the Prometheus Operator

What is a K8s Operator?

What is a K8s Operator?



Application specific
operational knowledge

What is a K8s Operator?



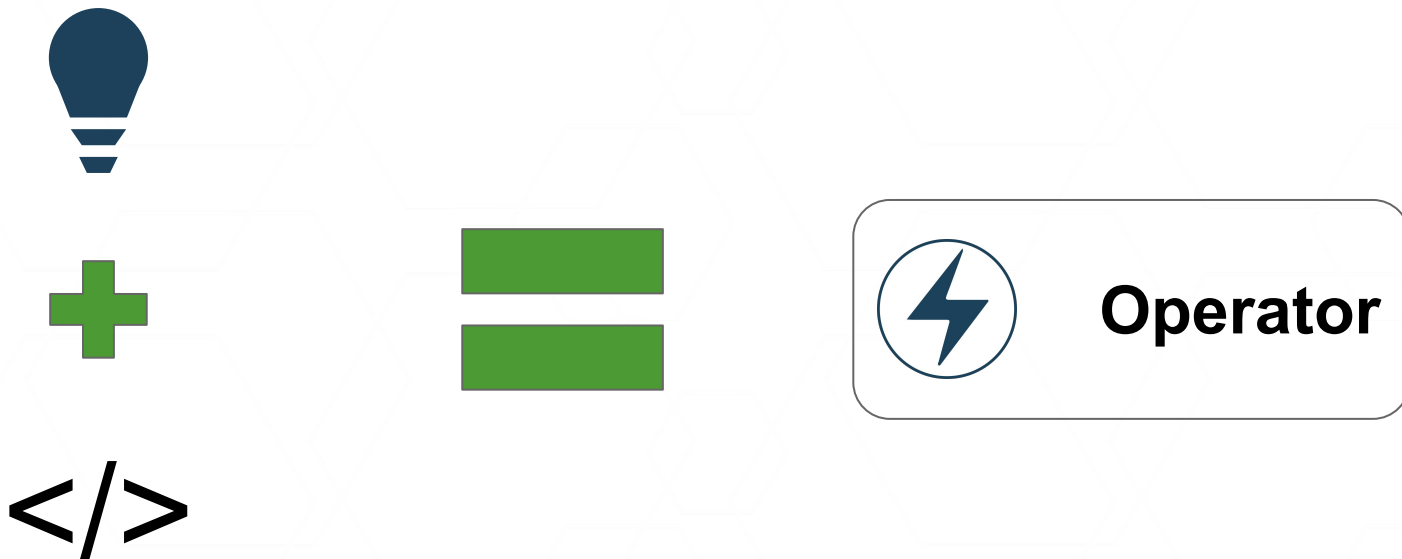
What is a K8s Operator?



What is a K8s Operator?



What is a K8s Operator?



Prometheus Operator

- Kubernetes native configuration
- Automated management and upgrades of Prometheus & Alertmanager

```
apiVersion: extensions/v1beta1
kind: Deployment
metadata:
  name: my-app
spec:
  ...
```

```
apiVersion: monitoring.coreos.com/v1alpha1
kind: Prometheus
metadata:
  name: prometheus-k8s
spec:
  ...
```

Kube-Prometheus

Single command to install:

- Prometheus & Alertmanager Cluster
- Alerting rules
- Dashboarding

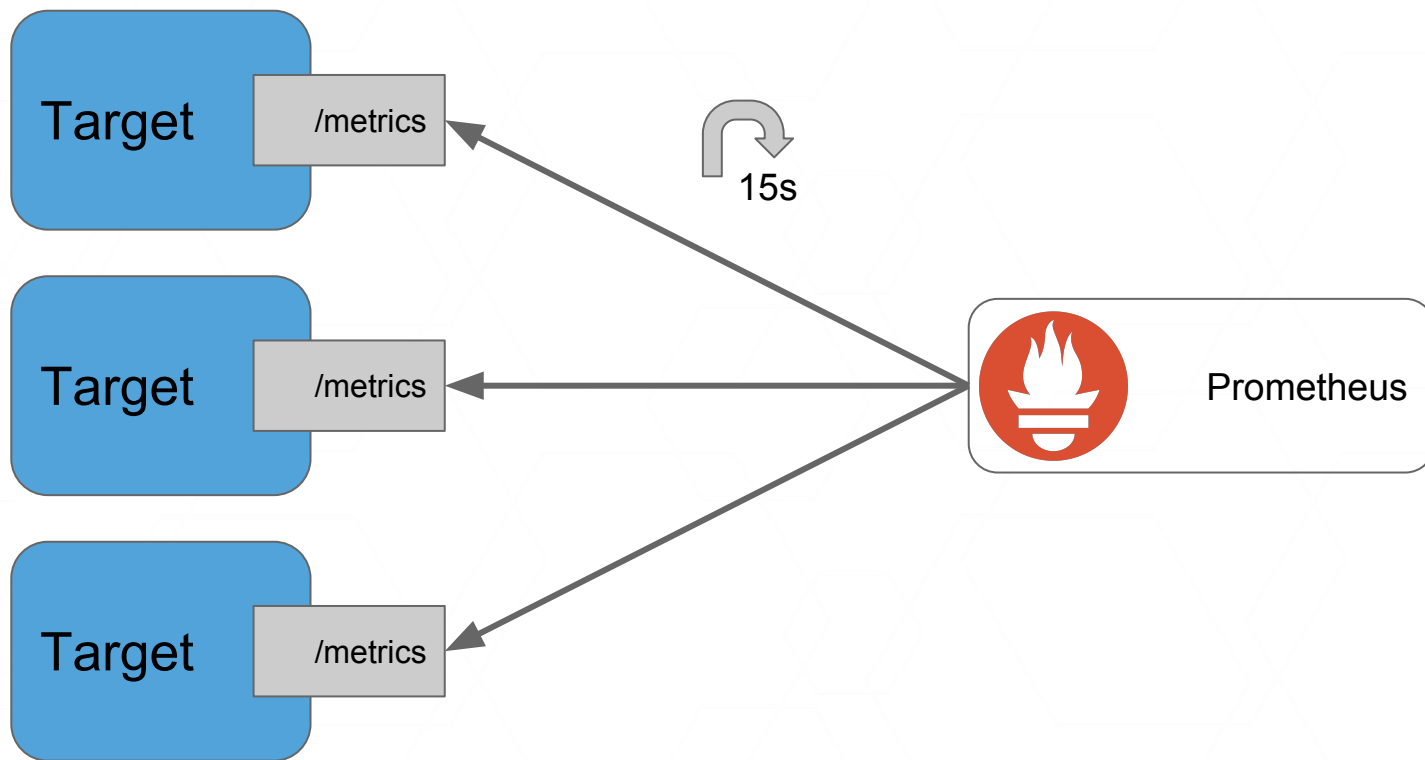
Demo

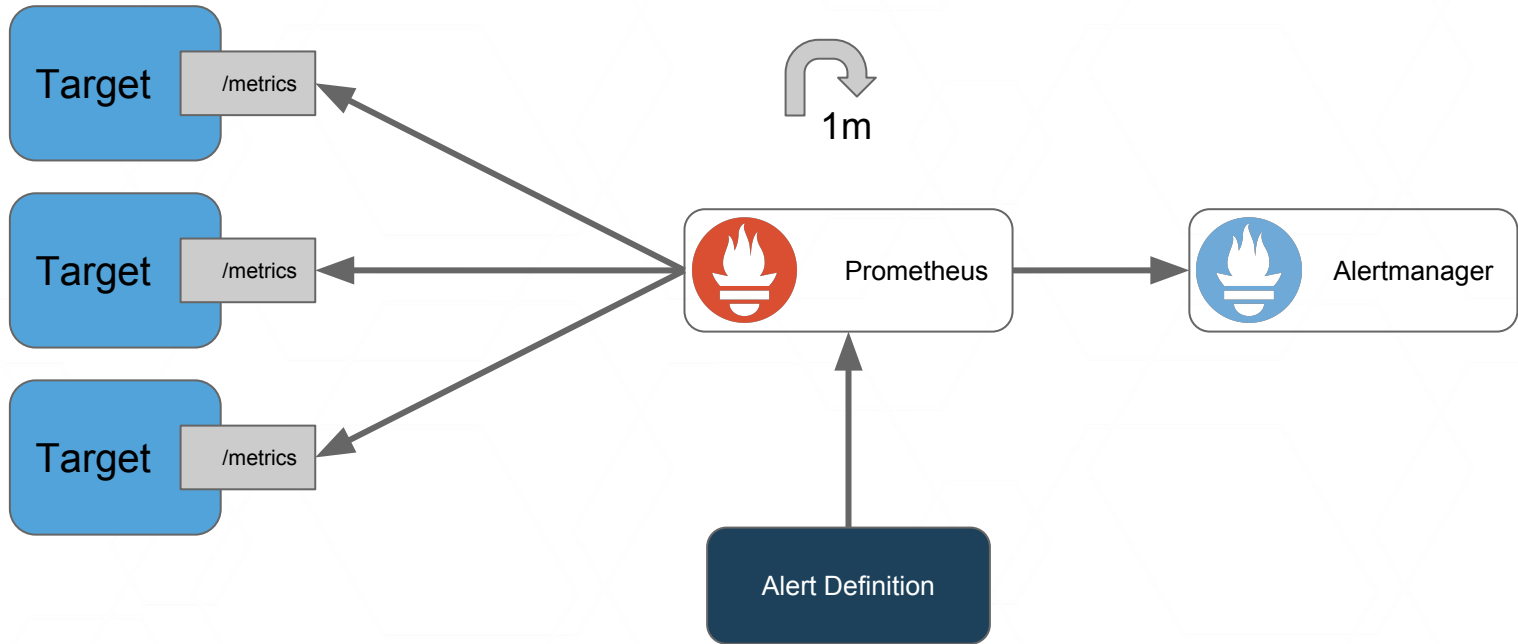
Recap

What is Prometheus?



- Pull-based
- Multi-Dimensional
- ***Metrics***, **not** logging, **not** tracing
- No magic!





Prometheus-Operator & Kube-Prometheus



Operator

Where to go from here?

Prometheus.io

 /coreos/prometheus-operator